VSEPR-SIM Theoretical Division

Internal Development Document — Continuation of Doc II

Inverse Kaluza-Klein Scale Symmetry

Document IV

Five-Dimensional Interval
and Scale-Projection Mechanics

5D logarithmic scale coordinate | Scale-dependent geometry |
Complex fiber bundle | Identity bundles at scale |
Projection operator formalism | Scale-dependent wavefunction |
Property activation functions | Cross-scale property map

Prerequisites: Doc II (Axioms A–J), Doc III (Kernel V4)

This document formalises the projection operator left open in Doc II Axiom B.4 and the dark-scale term χ_w in Doc II Eq. (F.5).

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§1 Motivation and Starting Point

Import block K *depends on Doc II Axioms B.2, B.3, B.4, F.5*

1.1 What ordinary spacetime misses

In special relativity an event is the four-vector

$$x^\mu = (ct,\, x,\, y,\, z), \tag{1}$$

and the spacetime interval is

$$ds_4^2 = -c^2\, dT^2 + dx^2 + dy^2 + dz^2. \tag{2}$$

Proper time accumulated along a worldline is

$$d\tau^2 = dT^2 - \frac{1}{c^2}(dx^2 + dy^2 + dz^2). \tag{3}$$

Standard result: a moving clock ages more slowly because spatial motion consumes what would otherwise be pure temporal evolution.

Extension hypothesis. Two objects may follow identical trajectories in the four visible coordinates yet accumulate different proper times if their internal *scale-state* evolves differently. This demands a fifth coordinate that encodes motion through resolution, internal phase structure, or scale-dependent access to degrees of freedom. It is not another place to stand; it is a record of which physical description level the system is occupying.

1.2 Why this addresses open variables from Doc II

Doc II left three items undefined (see Doc II Open Variable Register, Import Block J):
- $\chi_w(w)$: the dark-scale metric correction in Eq. (F.5).
- Π_n: the packing operator at each scale (Axiom B.3).
- $K_G(w) - K_{\mathrm{EM}}(w)$: the kernel difference in Eqs. (E.3)–(E.5).

This document provides explicit geometric machinery for all three by constructing a fifth coordinate from the physical scale variable ℓ and formalising the projection operator $\Pi_{m\leftarrow n}$ introduced abstractly in Doc II Axiom B.4.

§2 The Five-Dimensional Interval

Import block L *standalone; extends Eq. (F.1) of Doc II*

2.1 Naive 5D extension

A minimal five-dimensional interval appends one extra coordinate u:

$$ds_5^2 = -c^2\, dT^2 + dr^2 + \eta_u\, du^2, \tag{4}$$

where $dr^2 = dx^2 + dy^2 + dz^2$ and η_u is a signature coefficient to be determined. This form is insufficient as stated: without specifying what u measures, it says nothing.

2.2 The scale coordinate

The fifth coordinate is identified with physical resolution scale:

$$u = u(\ell), \qquad \ell \in [\ell_{\text{Planck}}, \ell_{\text{cosmo}}]. \tag{5}$$

The natural map is logarithmic, motivated by the fact that scale transitions span many orders of magnitude and the relevant structure at each level is set by ratios, not absolute differences:

$$\boxed{u(\ell) = \ln\left(\frac{\ell}{\ell_0}\right)} \tag{6}$$

where ℓ_0 is a fixed reference length. Consequently,

$$du = \frac{d\ell}{\ell}. \tag{7}$$

Reference scales appearing in the theory:

Level	Symbol	Approximate value
Planck	ℓ_{Planck}	$\sim 10^{-35}$ m
Nuclear	ℓ_{nuc}	$\sim 10^{-15}$ m
Atomic	ℓ_{atom}	$\sim 10^{-10}$ m
Molecular	ℓ_{mol}	$\sim 10^{-9}$ to 10^{-7} m
Macroscopic	ℓ_{mac}	$\gtrsim 10^{-6}$ m
Cosmological	ℓ_{cosmo}	$\gtrsim 10^{20}$ m

2.3 The physical 5D interval

Substituting (6) into (4):

$$\boxed{ds_5^2 = -c^2\, dT^2 + dr^2 + \eta_u \left(\frac{d\ell}{\ell}\right)^2} \tag{8}$$

The fifth term now explicitly measures the fractional change in resolution scale. A system that moves in scale (e.g. a particle undergoing a phase transition between energy regimes) contributes to ds_5^2 through $(d\ell/\ell)^2$, independently of its 3D spatial displacement.

Remark 2.1. Equation (8) reduces to the standard 4D interval (2) when $d\ell = 0$, i.e. when the description scale does not change. This is the correct classical limit.

2.4 Connection to the Doc II proper-time extension

Doc II Eq. (F.5) introduced an effective proper time

$$d\tau_{\text{eff}} = dt\sqrt{1 - \frac{v^2}{c^2} + \frac{2\Phi_G}{c^2} + \chi_w(w)}. \tag{F.5}$$

The dark-scale correction $\chi_w(w)$ is now identified. From (8), the proper-time element along a 5D worldline is

$$d\tau_5^2 = dT^2 - \frac{1}{c^2}\, dr^2 - \frac{\eta_u}{c^2}\left(\frac{d\ell}{\ell}\right)^2. \tag{9}$$

Comparing with (F.5):

$$\chi_w \;\longleftrightarrow\; -\frac{\eta_u}{c^2}\left(\frac{d\ell}{\ell\, dT}\right)^2. \tag{10}$$

The sign of χ_w is determined by η_u. For a spacelike scale-axis ($\eta_u > 0$), motion in scale *reduces* proper time, analogously to spatial velocity. The metric coefficient η_u is a free parameter of the theory until constrained by experiment; it encodes the coupling between scale-motion and physical aging.

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§3 Scale-Dependent Geometry

Import block M *standalone; prerequisite for §4*

3.1 Effective dimension

Standard geometry fixes spacetime dimension at $D = 4$ for all scales. The IKK framework replaces this with a scale-dependent effective dimension:

$$D_{\text{eff}} = D_{\text{eff}}(\ell). \tag{11}$$

The interpretation is operational: $D_{\text{eff}}(\ell)$ counts the number of physically accessible and dynamically active degrees of freedom at resolution scale ℓ.

At sub-nuclear scales, internal degrees of freedom (charge, colour, helicity) dominate spatial ones. At macroscopic scales, the three spatial dimensions fully activate and internal structure averages out. $D_{\text{eff}}(\ell)$ interpolates this behaviour continuously.

3.2 Scale-modulated metric

The most general scale-dependent interval consistent with the above writes scale-modulated coefficients on each axis:

$$ds^2 = -c^2\,dT^2 + a_1(\ell)\,dx^2 + a_2(\ell)\,dy^2 + a_3(\ell)\,dz^2 + a_u(\ell)\,du^2. \tag{12}$$

The coefficient $a_i(\ell) \in [0, 1]$ measures how fully active spatial axis i is at scale ℓ.

Definition 3.1 (Effective dimension from metric coefficients)**.**

$$D_{\text{eff}}(\ell) = \sum_{i=1}^{3} a_i(\ell) + a_u(\ell). \tag{13}$$

Limiting behaviours:
- Sub-nuclear ($\ell \lesssim 10^{-15}$ m): $a_1 \approx a_2 \gg a_3$ may describe a system where two internal-plane directions dominate over the stabilised third spatial axis.
- Chemical / material ($\ell \sim 10^{-10}$ to 10^{-7} m): $a_1 \approx a_2 \approx a_3 \approx 1$ and ordinary 3D geometry is fully recovered.
- Cosmological ($\ell \gtrsim 10^{20}$ m): $a_u(\ell) \gg 0$, meaning the scale-axis becomes dynamically relevant again through large-scale structure formation.

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§4 Complex Fiber Bundle

Import block N *standalone; required for §5*

4.1 Attaching a complex plane to every spacetime point

A physical point is usually specified as $p = (x, y, z, t)$. The extended structure attaches a local complex plane at each point:

$$p \;\mapsto\; (p,\, \mathbb{C}_p), \tag{14}$$

where $\mathbb{C}_p$ is the complex fibre above p.

This gives the structure of a fibre bundle

$$\pi : E \to M, \tag{15}$$

where M is the base scale-spacetime manifold defined by the interval (8), E is the total state space, and $\pi^{-1}(p) = \mathbb{C}_p$.

4.2 Local complex state

The local state in the fibre above p is written in polar form:

$$\psi(p) = A(p)\, e^{i\theta(p)}, \tag{16}$$

where $A(p)$ is the amplitude and $\theta(p)$ is the local phase.

Definition 4.1 (Wave behaviour as fibre projection)**.** Wave behaviour is the observed projection of a deeper, scale-dependent particle identity through the complex fibres. The wave is not the ontologically primary object. It is the projection signature of an identity-bearing particle state onto the accessible measurement basis.

This statement is the point at which the IKK framework diverges from “all is waves” interpretations. The priority is: *localised identity first, wave appearance second.*

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§5 Identity Bundles at Scale

Import block O *extends Doc II Axiom B.2 and Import Block C (packing chain)*

5.1 Scale levels (Doc II recap)

From Doc II Axiom B.2, a scale level is the triple

$$S_n = (D_n,\, M_n,\, L_n), \tag{17}$$

where D_n is the degree of freedom gained at scale n, M_n is the measurement basis, and L_n is the characteristic length scale.

5.2 Identity object at scale

A physical *identity object* at scale S_n is not merely a point. It is the structured quintuple

$$\boxed{P_n = (x^\mu,\; \mathbb{C}_p,\; D_n,\; M_n,\; L_n)} \tag{18}$$

The components are:

- x^μ: spacetime coordinates in the base manifold M.
- $\mathbb{C}_p$: attached complex fibre encoding local phase and amplitude.

- D_n: distinguishability content — how much identity information is recoverable at scale n.
- M_n: the measurement basis available at scale n.
- L_n: the length scale.

Remark 5.1. Equation (18) extends the Doc II Python/C++ objects `ScaleEntry` and `IdentityBundle` by adding the complex-fibre slot $\mathbb{C}_p$. The packing operator Π_n from Doc II Axiom B.3 maps $P_n \to P_{n+1}$ with compression of D_n.

5.3 Identity persistence across scales

Identities at lower scales do not vanish when a higher-scale identity assembles from them. They remain present; the measurement system can only access their projected contribution.

Proposition 5.1 (Scale coexistence)**.** *For $m > n$ (coarser > finer):*

$$P_n \text{ exists} \implies P_m = \Pi_{m\leftarrow n}(P_n) \text{ coexists with } P_n. \tag{19}$$

Projection determines visibility*, not* existence*.*

This avoids the emergence-as-deletion error: a molecule's formation does not erase its constituent electrons; it hides their individual projections behind the molecular-scale observable.

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§6 Projection Operator Formalism

Import block P *formalises Doc II Axiom B.4; connects to Axiom B.5 residual*

6.1 Definition

Definition 6.1 (Scale projection operator)**.** The projection operator from scale S_n to scale S_m (with $m \geq n$, i.e. coarser $\geq$ finer) is

$$\Pi_{m\leftarrow n} : P_n \longrightarrow P_m^{\mathrm{obs}}, \tag{20}$$

or, in state-vector notation,

$$|\Psi_m\rangle = \Pi_{m\leftarrow n} |\Psi_n\rangle. \tag{21}$$

6.2 Distinguishability loss

Projection is not copying. It loses, compresses, averages, or re-expresses information. Define the *distinguishability* D_n as the amount of identity-resolving information at scale S_n. Projection to a coarser scale satisfies

$$D_m \leq D_n, \qquad m \geq n. \tag{22}$$

The lost content is

$$\Delta D_{n\to m} = D_n - D_m \geq 0. \tag{23}$$

$\Delta D_{n\to m}$ is the formal account of what appears at scale S_m as: uncertainty, phase spread, probability amplitude, quantum fuzziness, decoherence, or unresolved wave-like behaviour.

6.3 Decomposition identity

The full state at scale S_n decomposes exactly as:

$$\boxed{|\Psi_n\rangle = \underbrace{\Pi_{m\leftarrow n}|\Psi_n\rangle}_{|\Psi_m^{\mathrm{obs}}\rangle} + \underbrace{(\mathbf{I} - \Pi_{m\leftarrow n})|\Psi_n\rangle}_{|\Psi_{n\backslash m}^{\mathrm{hid}}\rangle}} \tag{24}$$

where:
- $|\Psi_m^{\mathrm{obs}}\rangle = \Pi_{m\leftarrow n}|\Psi_n\rangle$ is the component visible at scale S_m.
- $|\Psi_{n\backslash m}^{\mathrm{hid}}\rangle = (\mathbf{I} - \Pi_{m\leftarrow n})|\Psi_n\rangle$ is the component hidden from scale S_m measurement.

Remark 6.1 (Bridge between matrix and wave mechanics)*.* Matrix mechanics describes the full transition structure $|\Psi(t)\rangle = e^{-iHt/\hbar}|\Psi(0)\rangle$. Wave mechanics describes the coordinate representation $\psi(x,t) = \langle x|\Psi(t)\rangle$. In the IKK framework (24) unifies these:

$$\text{Matrix state:} \quad |\Psi_n\rangle = \text{full transition structure at scale } S_n. \tag{25}$$

$$\text{Wavefunction:} \quad \psi(x,t) = \text{projected coordinate-scale representation.} \tag{26}$$

The wavefunction is not primary. It is the observed projection of unresolved identity content onto the position basis at a given scale.

6.4 Length-dependent projection

Making the projection explicitly scale-dependent:

$$\Pi_{m\leftarrow n} = \Pi(L_m,\, L_n) \tag{27}$$

so that the observed state at resolution L_m given a true state at L_n is

$$|\Psi(L_m)\rangle = \Pi(L_m, L_n)\,|\Psi(L_n)\rangle. \tag{28}$$

For a fundamental identity, the residual hidden from all macroscopic measurement is

$$|\Psi^{\perp}(L)\rangle = (\mathbf{I} - \Pi_{\ell})\,|\Psi_{\mathrm{true}}\rangle, \tag{29}$$

where Π_ℓ is the projector associated with measurement at length scale L.

6.5 Connection to the Doc II residual axiom

Doc II Axiom B.5 defined the residual

$$\mathcal{R}_S = \mathcal{U}_S(\mathcal{I}) - \mathcal{E}_S(\mathcal{O}_S). \tag{B.5}$$

This is now the projection residual of (29):

$$\mathcal{R}_S \;\equiv\; |\Psi^{\perp}(L_S)\rangle = (\mathbf{I} - \Pi_{L_S})\,|\Psi_{\mathrm{true}}\rangle. \tag{30}$$

The residual is zero only when projection is injective at scale S, consistent with Axiom B.5.

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§7 Scale-Dependent Wavefunction

Import block Q *depends on §4, §6*

7.1 Standard form and its reinterpretation

The standard wavefunction $\psi(x,t) = \langle x|\Psi(t)\rangle$ is the projection of a state vector onto the position basis. In the IKK framework this is promoted to a *scale-dependent* projection:

$$\boxed{\psi(x,t;\,L) = \langle x, L\,|\,\Psi_{\text{true}}\rangle = \Pi_{x,L}\,|\Psi_{\text{true}}\rangle} \tag{31}$$

The wavefunction is now the position projection at a specific length scale L. Two observers using different resolution scales $L_1 \neq L_2$ will obtain different wavefunctions for the same underlying identity state.

7.2 Born rule reinterpretation

The probability density becomes:

$$|\psi(x,t;\,L)|^2 = \text{density of projected identity-access at } (x,t) \text{ with resolution } L. \tag{32}$$

This is *not* a statement that the particle "is somewhere because the universe enjoys gambling." It is a statement about how much of the deeper identity structure is accessible through a measurement basis of resolution L.

7.3 Wave-particle duality as scale-projection mismatch

Proposition 7.1 (Resolution of wave-particle duality)**.** *Wave-particle duality is not a fundamental ontological duality. It is a* scale-projection mismatch*: the observer's resolution length L_{obs} does not fully resolve the identity length L_n of the object being measured. Specifically:*

$$L_{\text{obs}} \gg L_n \implies \Delta D_{n\to\text{obs}} \gg 0 \implies \textit{wave-like appearance.} \tag{33}$$

Point-like (particle) behaviour is recovered when $L_{\text{obs}} \lesssim L_n$ and the projection is nearly injective.

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§8 The Scale Ladder

Import block R *extends Doc II Scale Ladder (Import Block B)*

8.1 Logarithmic scale spacing

The scale ladder from Doc II is

$$\cdots \to S_{-2} \to S_{-1} \to S_0 \to S_1 \to S_2 \to S_3 \to \cdots \tag{34}$$

with logarithmically spaced length scales:

$$L_n = L_0\,\lambda^n, \tag{35}$$

where λ is the scale multiplier. For decade spacing $\lambda = 10$:

$$L_n = L_0 \cdot 10^n. \tag{36}$$

The scale coordinate of rung n is

$$u_n = \ln\!\left(\frac{L_n}{L_0}\right) = n\ln\lambda. \tag{37}$$

8.2 Projection between rungs

The projection from rung n to rung m (with $m > n$, i.e. upward toward coarser description) is

$$|\Psi_m\rangle = \Pi(L_m, L_n)\,|\Psi_n\rangle = \Pi(L_0\lambda^m,\, L_0\lambda^n)\,|\Psi_n\rangle. \tag{38}$$

The distinguishability loss across $k = m - n$ rungs is

$$\Delta D_{n\to n+k} = D_n - D_{n+k}. \tag{39}$$

In a uniform ladder model one may write $D_n - D_{n+1} = \delta D$ (constant per rung), giving $\Delta D_{n\to n+k} = k\,\delta D$. The actual rung-to-rung information loss is a free parameter to be fitted from data at each scale transition.

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§9 Property Activation Functions

Import block S *depends on §8*

9.1 Activation function definition

A physical property Q is not equally accessible at all scales. Define the *activation function*

$$A_Q(\ell) \in [0, 1], \tag{40}$$

where $A_Q(\ell) = 0$ means the property is inaccessible or undefined at scale ℓ, and $A_Q(\ell) = 1$ means it is fully expressed.

A smooth sigmoid form consistent with the logarithmic scale coordinate is:

$$\boxed{A_Q(\ell) = \frac{1}{1 + \exp\!\left[-k_Q\,(\ln\ell - \ln L_Q)\right]}} \tag{41}$$

Parameters:
- L_Q: the characteristic length at which the property activates (inflection point of the sigmoid in log-scale).
- $k_Q > 0$: sharpness of the transition. Large k_Q approximates a step function; small k_Q gives a smooth crossover spanning many decades.

9.2 Observed property

The observed value of property Q at scale ℓ is

$$Q_{\text{obs}}(\ell) = A_Q(\ell)\,\Pi_\ell(Q_{\text{true}}), \tag{42}$$

where Π_ℓ is the projection operator at scale ℓ (see (31)).

9.3 Physical examples of activation

- **Spin.** Spin belongs to internal/gauge-scale identity. It always exists in the identity structure but activates magnetically only when projected to electromagnetic scales:

$$\mu_{\text{mag}} = \Pi_{\text{macro}\leftarrow\text{spin}}(s), \quad A_{\text{spin}}(\ell_{\text{macro}}) \approx 1 \text{ (via spin–orbit coupling).} \tag{43}$$

- **Temperature.** Temperature is undefined for a single particle. It activates across the statistical ensemble transition:

$$k_B T \sim \frac{2}{f(\ell)} \langle E_{\mathrm{kin}} \rangle, \tag{44}$$

where $f(\ell) = D_{\mathrm{eff}}(\ell)$ is the number of active degrees of freedom at scale ℓ. This directly connects the activation function to thermodynamic behaviour: $f(\ell)$ varies from 3 (monatomic) to higher values as internal modes activate at larger scales.
- **Chemical bond geometry.** Bond angles are undefined at sub-nuclear scales. The activation length is $L_Q \sim 10^{-10}$ to 10^{-9} m:

$$\theta_{ijk} = \Pi_{\mathrm{geometry}}(\psi_i, \psi_j, \psi_k), \quad A_{\mathrm{bond}}(\ell) \approx 0 \ \text{ for } \ \ell \ll 10^{-10}\,\mathrm{m}. \tag{45}$$

- **Inertial mass.** Mass is the projected resistance of identity to change of motion state:

$$m_{\mathrm{obs}} = \Pi_{\mathrm{inertial}}(P_{\mathrm{true}}), \tag{46}$$

and for a composite body

$$M_{\mathrm{obs}}\, c^2 = \sum_i m_i\, c^2 + E_{\mathrm{binding}} + E_{\mathrm{field}}. \tag{47}$$

Mass is a projected property of the full identity system, not a simple sum of component tags.
- **Dark-matter-like gravitational residuals.** From the Doc II kernel (Eq. E.3), the gravitational observable at galactic scale is

$$G_{\mathrm{obs}} = \Pi_{\mathrm{galactic}}(P_{\mathrm{matter}}) + \Delta G_{\mathrm{scale}}, \tag{48}$$

where $\Delta G_{\mathrm{scale}}$ is the residual from hidden scale-coordinate structure. The galactic-scale projection operator may expose gravitational behaviour that is absent at smaller scales. This reproduces the Doc II dark-matter kernel structure: the integrand $[K_G(w) - K_{\mathrm{EM}}(w)]$ in Eq. (E.3) is now interpretable as the difference between the scale-integrated activation of the gravitational projection and the EM projection.

$$K_G(w) - K_{\mathrm{EM}}(w) \ \equiv \ A_G(e^w \ell_0) - A_{\mathrm{EM}}(e^w \ell_0). \tag{49}$$

This is a hypothesis, not a derivation. Numerical confirmation requires fitting k_G, k_{EM}, L_G, L_{EM} to rotation curve data.

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§10 Cross-Scale Property Map

Import block T *summary table; depends on §9*

A property Q is assigned the quadruple

$$Q \ \mapsto \ (D_Q,\, L_Q,\, M_Q,\, \Pi_Q) \tag{50}$$

where D_Q is the effective dimension required, L_Q the activation length, M_Q the measurement basis, and Π_Q the projection operator that makes the property accessible.

Scale regime	**Length**	**Dominant identity**	**Effective D**	**Projection behaviour**
Primitive / Planck	10^{-35} m	Pre-geometric identity	Unknown / hidden	Fully hidden; only existence gate active
Gauge / sub-particle	10^{-18}–10^{-15} m	Quark, lepton, gauge	Internal + low spatial	Appears as quantum number state; $\lvert\psi\rangle$ dominated by internal channels
Nuclear	10^{-15}–10^{-14} m	Nucleon / nucleus	3D onset + internal binding	Particle identity partially stabilised; binding energy dominant
Atomic	10^{-10} m	Atom	3D + phase shell	Wavefunction dominates observable; orbital structure projects
Molecular	10^{-9}–10^{-7} m	Molecule	3D structure	Internal phase projects into bond geometry; chirality activates
Mesoscopic / material	10^{-6}–10^{-3} m	Grain, domain, defect field	3D continuum onset	Many identities average into continuum fields; stress, diffusion activate
Macroscopic	10^{-3}–10^{3} m	Body / object	Classical 3D	Quantum identity mostly hidden; force, torque, heat accessible
Planetary	10^{6}–10^{8} m	Planet / system	3D + gravitational field	Internal matter projected as mass; orbital mechanics dominant
Stellar / galactic	10^{9}–10^{21} m	Star / galaxy	3D + scale-gravity	Scale-axis residuals may reappear; rotation-curve anomalies possible
Cosmological	$\gtrsim 10^{21}$ m	Universe-scale identity	4D / 5D effective	Scale-axis dominant; expansion and large-scale structure driven by $a_u(\ell)$

Table 1: Cross-scale property map: identity, dimension, and projection behaviour across fourteen orders of magnitude.

§11 Updated Open Variable Register

Import block U — *updates Doc II Import Block J*

Variable	Status after Doc IV	Remaining requirement
$\chi_w(w)$	Identified via (10)	Value of η_u needs experimental or theoretical constraint.
$K_G(w)$, $K_{\text{EM}}(w)$	Identified as activation functions (49)	Parameters k_G, k_{EM}, L_G, L_{EM} need fitting.
Π_n	Formalised as $\Pi_{m\leftarrow n}$ in (20)–(24)	Explicit matrix form for each scale transition still needed.
$D_{\text{eff}}(\ell)$	Defined via metric coefficients (13)	Functional form of $a_i(\ell)$ curves needs derivation or data fit.
$\hat{H}_w$	Not addressed in this document	Dark-scale Hamiltonian remains undefined. Required for QM predictions.
$\mathbf{T}_{AB}$	Unchanged from Doc II	Relation-particle dynamics still undefined.
$\mathbf{W}$	Unchanged from Doc II	Hidden-intensity weight matrix needs specific form per channel.
k_Q, L_Q	New (this document)	Sigmoid parameters for each property require calibration against experimental activation energies and length-scale data.

§12 Non-Technical Summary

In ordinary relativity a moving clock runs slowly because spatial motion diverts what would otherwise be pure temporal evolution. This document extends that idea by proposing that physical systems may also move through *scale*: the same object described at different resolutions occupies different rungs of a logarithmic scale ladder, and this motion contributes to the proper time integral through the fifth term $\eta_u(d\ell/\ell)^2$ in the interval (8).

Every physical point carries an attached complex plane, making phase and amplitude geometric features present at all scales. What changes from one scale to the next is not whether quantum structure exists, but how much of it can be projected into the observer's measurement basis at that resolution. The projection operator $\Pi_{m\leftarrow n}$ maps the full identity-bearing state at a fine scale into the observable state at a coarser one. The complement — the hidden component $|\Psi^{\text{hid}}_{n\setminus m}\rangle$ in (24) — appears at the coarser scale as probability spread, phase uncertainty, or wave-like behaviour.

In this view, wave-particle duality is not a fundamental fact about matter. It is a scale-projection mismatch: when the resolution of the measurement does not reach the characteristic length of the identity object, the missing distinguishability content projects as a wave envelope.

Property activation functions (41) formalise the experimental observation that certain properties — temperature, bond geometry, stress, dark-matter-like gravitational residuals

— are meaningless below their characteristic activation scale but fully real above it. The five-dimensional interval (8) provides the geometric spine on which all of this sits.

IKK Doc IV — VSEPR-SIM Theoretical Division. The fifth dimension was here all along; it just needed a better address.